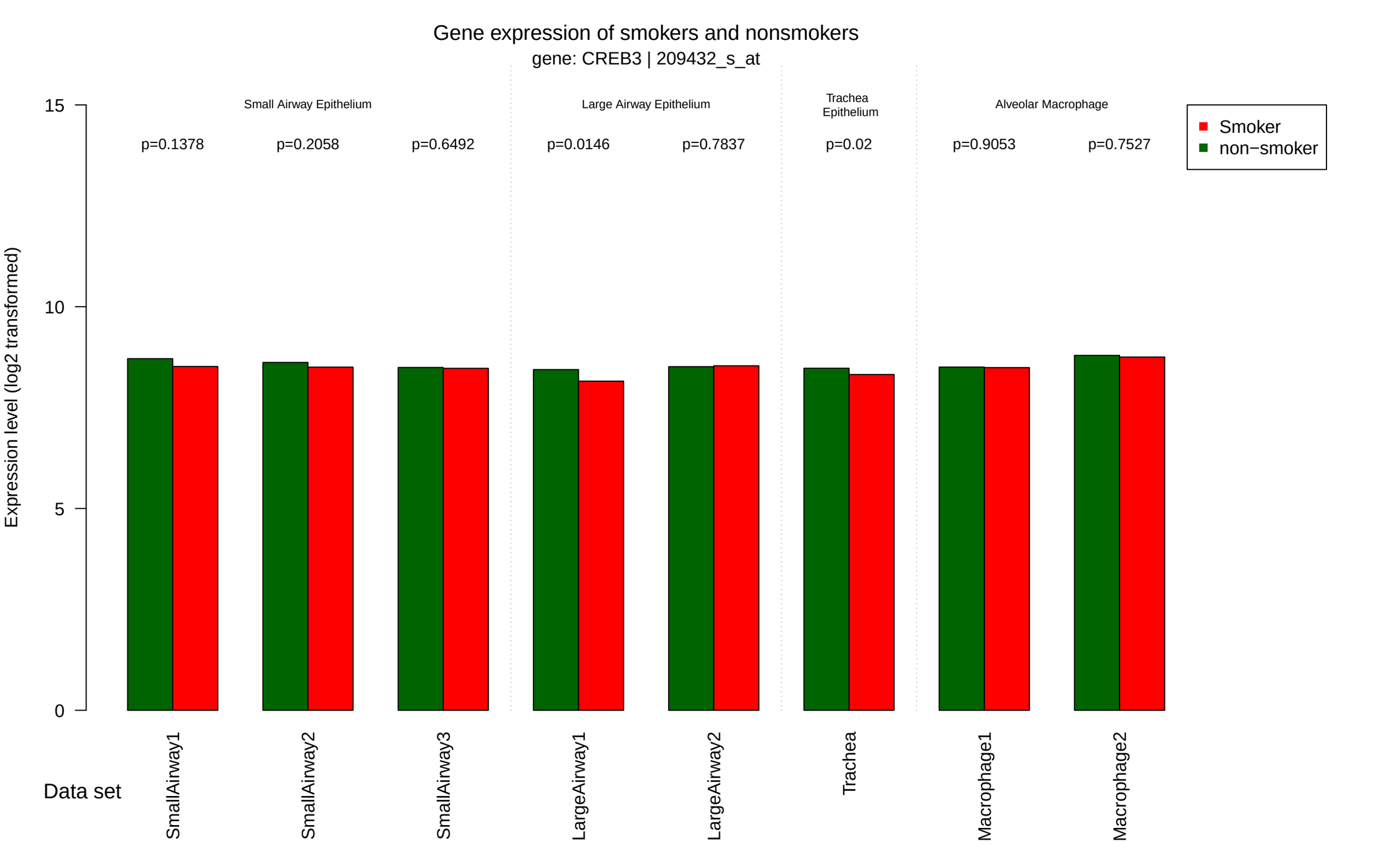
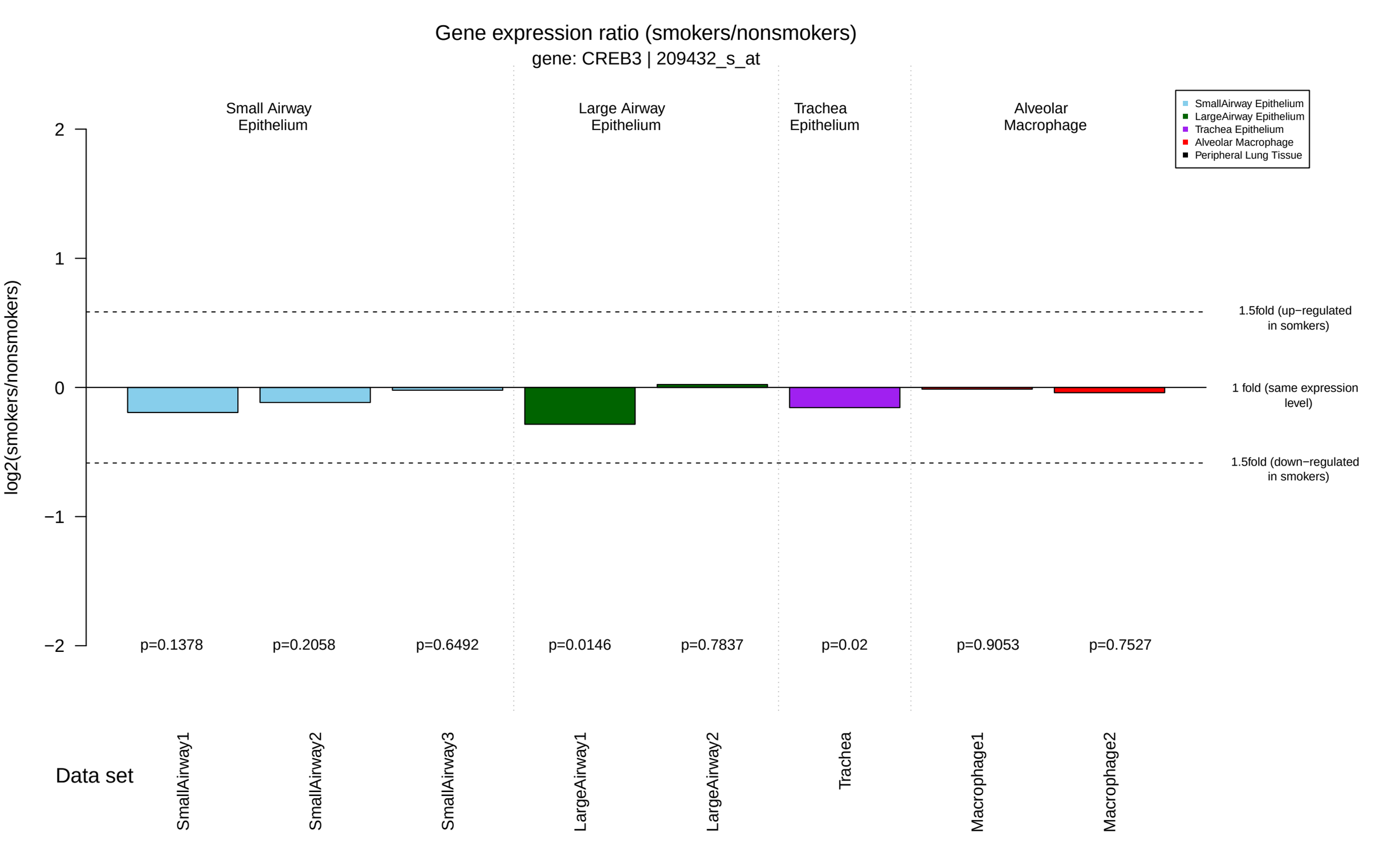
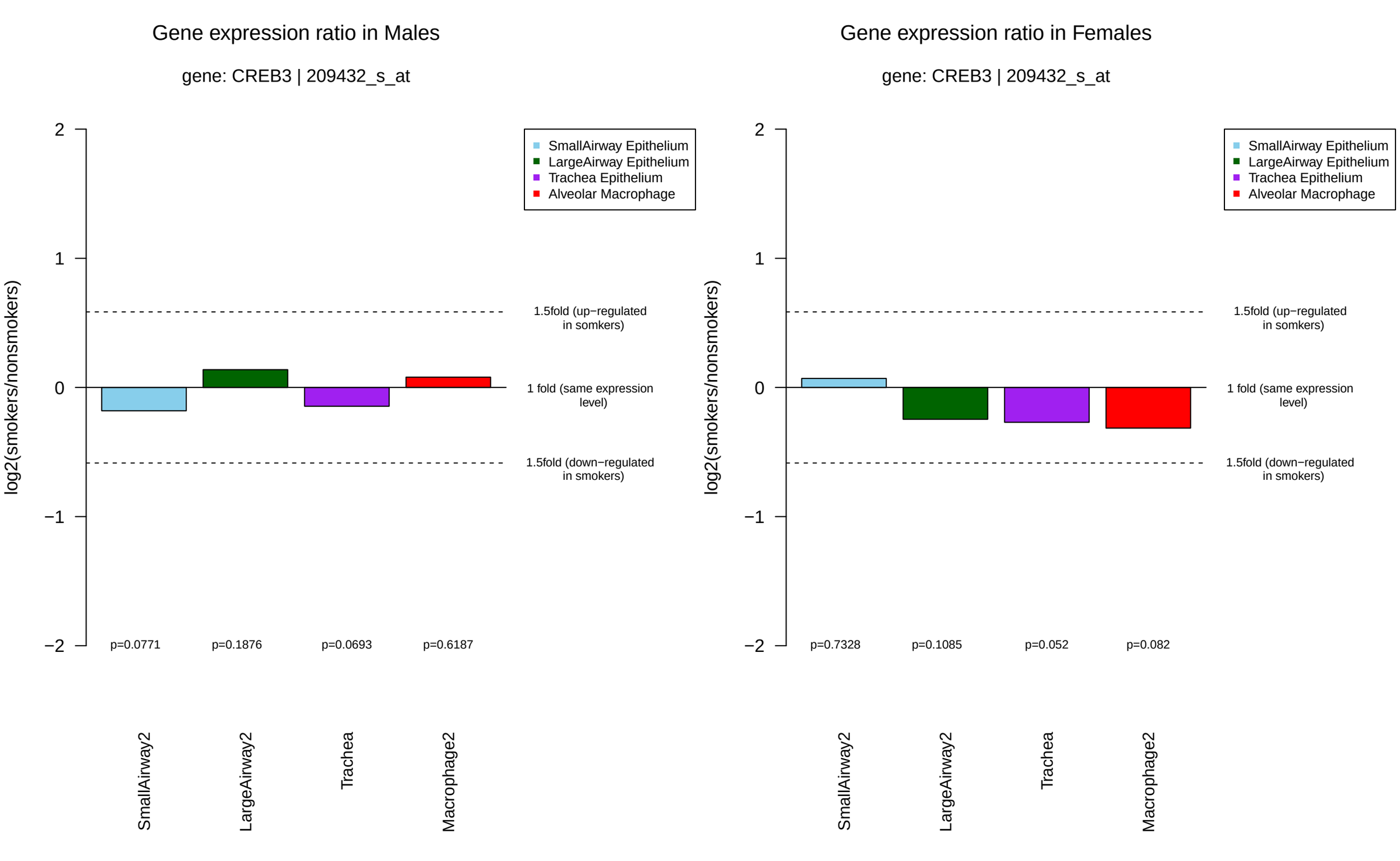
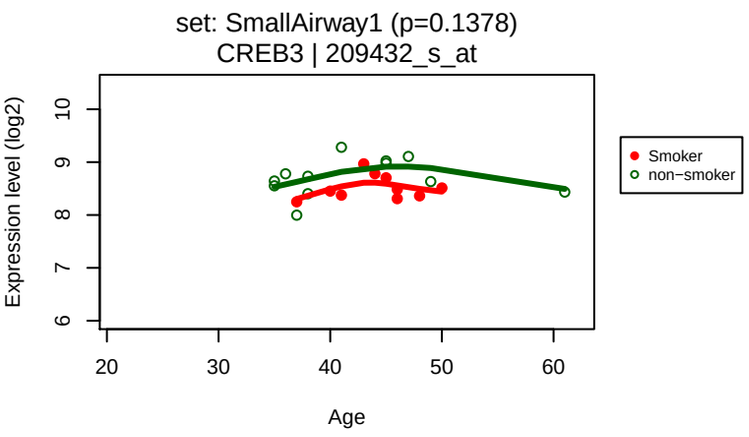
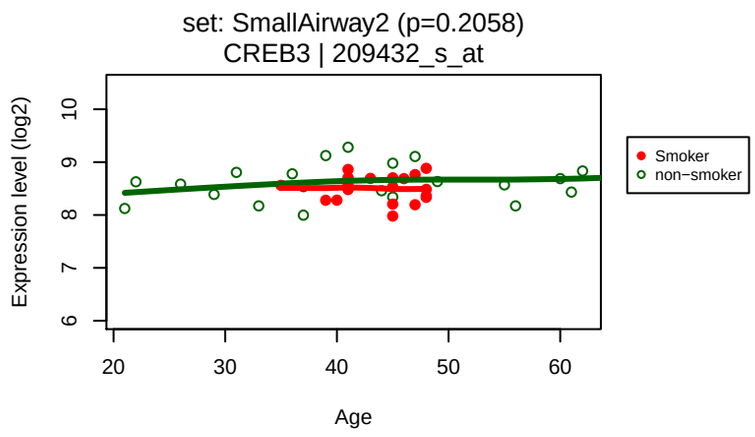
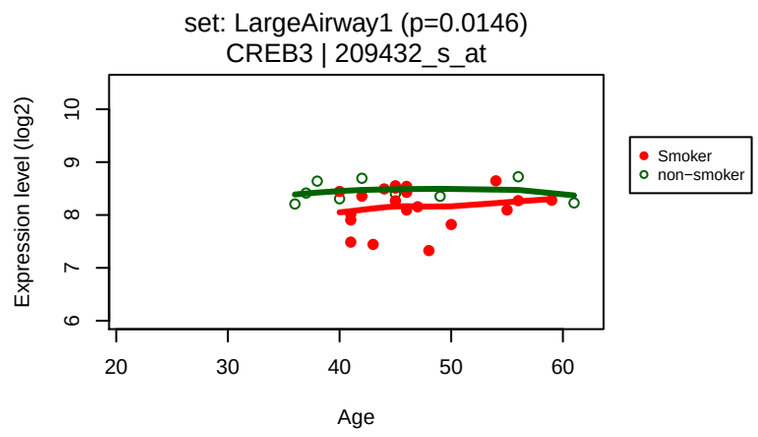

Gene expression of smokers and nonsmokers
gene: CREB3 | 209432_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1378
p=0.2058
p=0.6492
p=0.0146
p=0.7837
p=0.02
p=0.9053
p=0.7527
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

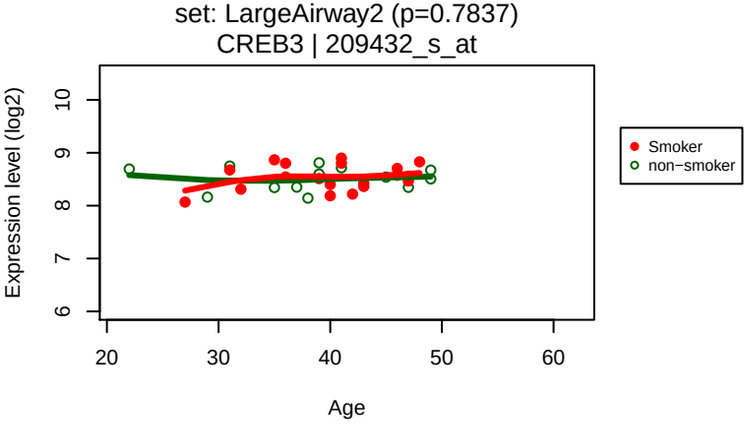

Gene expression ratio (smokers/nonsmokers)
gene: CREB3 | 209432_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1378
p=0.2058
p=0.6492
p=0.0146
p=0.7837
p=0.02
p=0.9053
p=0.7527
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

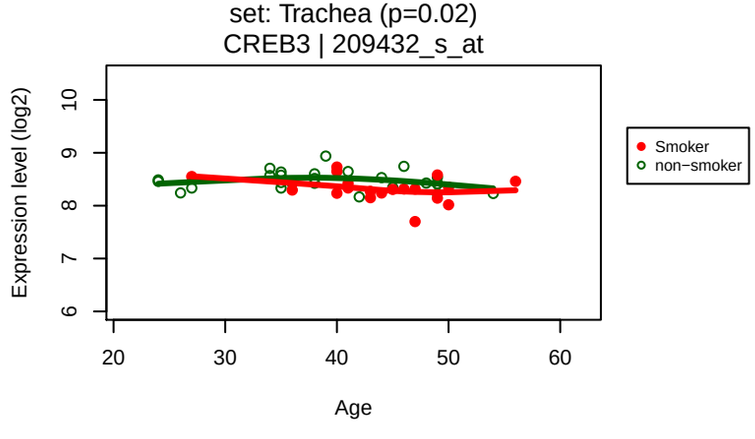

Gene expression ratio in Males
gene: CREB3 | 209432_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0771
p=0.1876
p=0.0693
p=0.6187
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB3 | 209432_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7328
p=0.1085
p=0.052
p=0.082
SmallAirway2
LargeAirway2
Trachea
Macrophage2

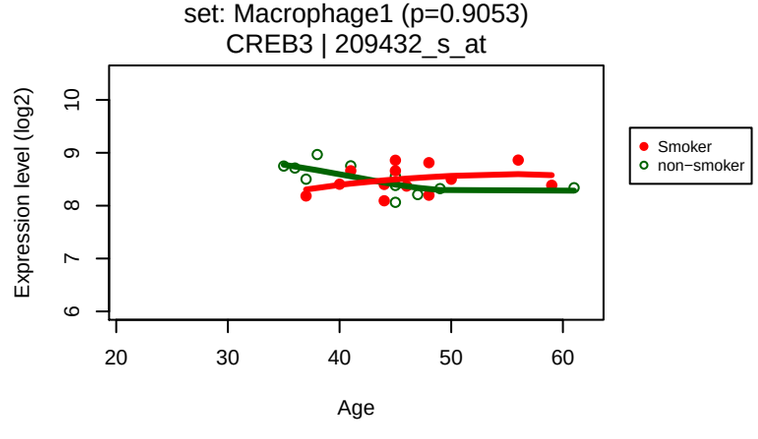

set: SmallAirway1 (p=0.1378)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker

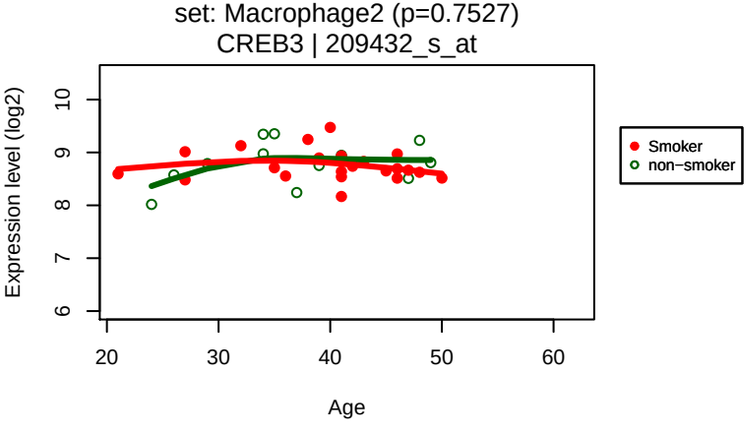

set: SmallAirway2 (p=0.2058)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker

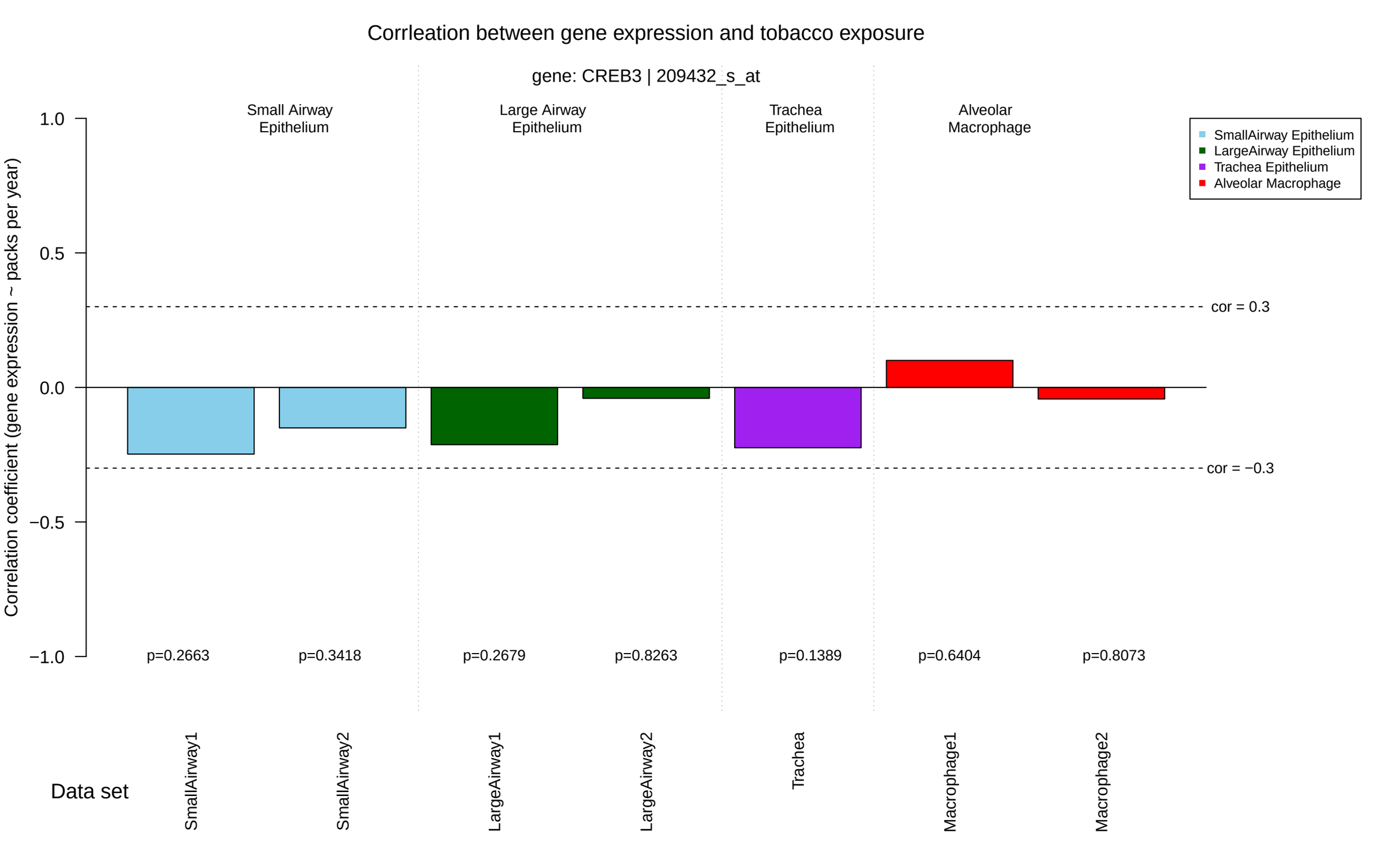

set: LargeAirway1 (p=0.0146)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7837)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.02)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9053)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7527)
CREB3 | 209432_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB3 | 209432_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2663
p=0.3418
p=0.2679
p=0.8263
p=0.1389
p=0.6404
p=0.8073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2